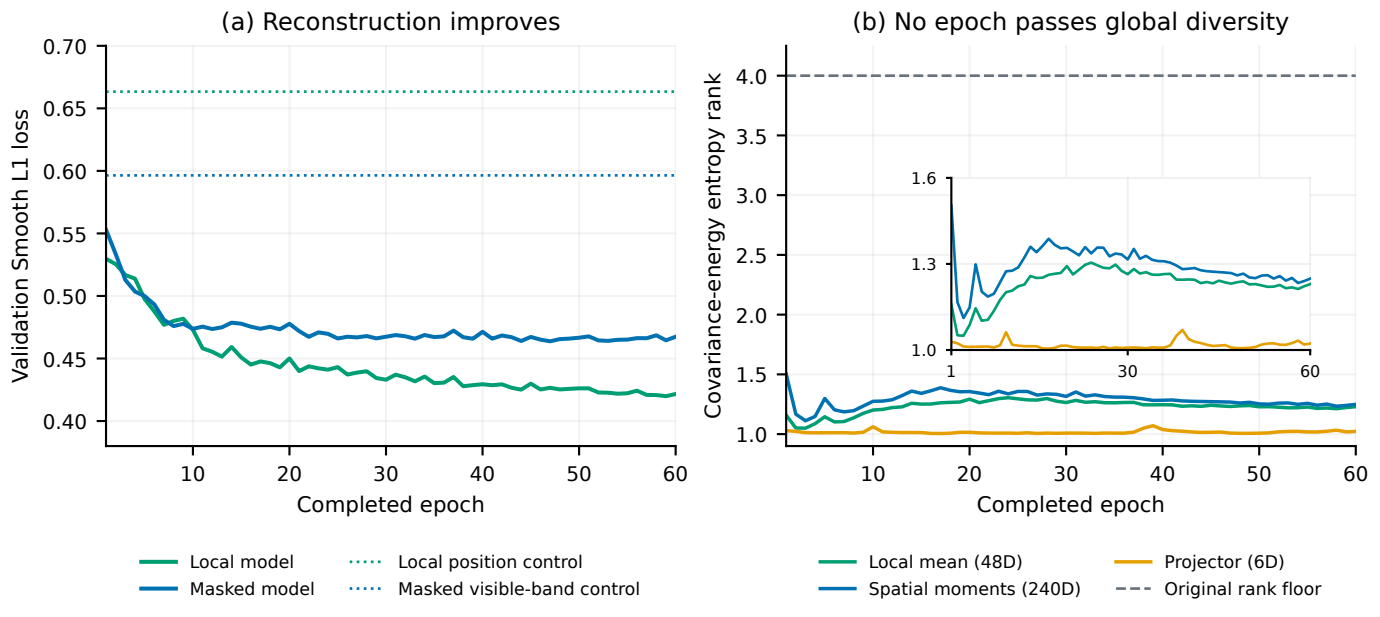
(a) Reconstruction improves
Validation Smooth L1 loss
0.70
0.65
0.60
0.55
0.50
0.45
0.40
10
20
30
40
50
60
Completed epoch
Local model
Masked model
Local position control
Masked visible-band control
(b) No epoch passes global diversity
Covariance-energy entropy rank
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.6
1.3
1.0
1
30
60
Completed epoch
Local mean (48D)
Spatial moments (240D)
Projector (6D)
Original rank floor